

SoleMate – E-Commerce Platform for Shoe Retail

Objective:

Design, develop, and deploy a fully functional e-commerce application for "SoleMate" – a shoe company aiming to expand its retail presence into the online marketplace. The application will provide an intuitive, secure, and responsive platform for customers to browse, purchase, and review products, with integrated inventory, order tracking, and analytics for business operations.

Business Context:

With increasing competition in the footwear industry and consumer preference for online shopping, SoleMate needs a robust e-commerce platform to cater to its customers across desktop and mobile devices. The platform must support scalability, secure transactions, seamless user experience, and integration with backend systems like inventory and CRM.

Scope of Work (Aligned to SDLC Phases)

1. Requirement Gathering & Analysis

Activities:

- Conduct stakeholder interviews (Sales, Marketing, Warehouse, IT teams) - Assume the details
- Define business goals and target audience
- Capture functional requirements:
 - User registration & authentication
 - Product catalog with filtering and search
 - Product detail pages with reviews and ratings
 - Shopping cart and checkout
 - Payment gateway integration (Stripe/PayPal/UPI)
 - Order tracking & history
 - Customer profile management
 - Admin dashboard for product and order management
 - Inventory management & stock alerts
 - Promotional features (discount codes, offers)

- Capture non-functional requirements:
 - Performance: Load <2 seconds
 - Scalability: Handle 50,000 concurrent users
 - Security: PCI-DSS compliance, data encryption
 - Compatibility: Mobile, tablet, and desktop
- Define acceptance criteria for each feature

Deliverables:

- SRS (Software Requirements Specification)
- Use case diagrams & user stories
- Requirements traceability matrix (RTM)

2. Planning & Estimation

Activities:

- Select tech stack:
 - **Frontend:** React.js, Next.js for SSR, TailwindCSS or any other technology
 - **Backend:** Node.js with Express or any other technology
 - **Database:** PostgreSQL for relational data, Redis for caching or any other technology
 - **Payment:** Stripe API or any other technology
 - **Cloud Deployment:** AWS (EC2, RDS, S3, CloudFront) or any other technology
- Create project plan with milestones
- Resource allocation (Frontend Devs, Backend Devs, QA, DevOps)
- Risk analysis and mitigation plan
- Budget estimation

Deliverables:

- Project plan with Gantt chart
- Resource allocation matrix
- Risk register
- Cost estimation report

3. System Design

Activities:

- **High-Level Architecture:**
 - MVC structure with API Gateway
 - Microservices for product, order, and payment services
 - CDN for static content delivery
- **Database Design:**
 - ER diagram for products, users, orders, payments, inventory
- **API Design:**
 - REST APIs for mobile and web
 - Swagger documentation
- **UI/UX Design:**
 - Wireframes for key screens (Figma)
 - Interactive prototypes
 - Accessibility compliance check (WCAG 2.1)

Deliverables:

- HLD & LLD documents
- ER diagram
- API documentation
- UI wireframes and prototypes

4. Development

Activities:

- Setup development environment & version control (GitHub/GitLab)
- Implement features in sprints (Agile methodology)
- Code reviews and merge approvals
- Integrate APIs with frontend components
- Unit testing for each module
- Build responsive layouts

Deliverables:

- Source code repository

- API endpoints with documentation
- Unit test reports

5. Testing

Activities:

- **Functional Testing:** Verify features against SRS
- **Integration Testing:** Ensure smooth interaction between modules
- **Performance Testing:** Load and stress tests
- **Security Testing:** SQL injection, XSS, CSRF checks
- **UAT (User Acceptance Testing):** Real-world usage simulation

Deliverables:

- Test plan & test cases
- Bug/defect reports in Jira
- UAT sign-off document

6. Deployment

Activities:

- Setup CI/CD pipelines with GitHub Actions/Jenkins
- Deploy backend to AWS EC2, frontend to AWS Amplify/CloudFront or other technologies
- Final production release

Deliverables:

- Deployment checklist
- Rollback plan

7. Maintenance & Support

Activities:

- Monitor application

- Fix post-release defects
- Add new features based on customer feedback (Optional / Assume the details)

Deliverables:

- Maintenance logs
- Performance reports
- Feature enhancement backlog

Project Evaluation Metrics

Area	Metric	Target
Requirement Quality	% of requirements covered in RTM	$\geq 100\%$
Design Quality	% of design changes after development start	$\leq 5\%$
Development Quality	Code coverage (unit tests)	$\geq 80\%$
Performance	Avg. page load time	$\leq 2s$
Security	Vulnerabilities detected post-launch	0 critical
Testing Efficiency	Defect detection ratio (pre vs. post-release)	$\geq 90\%$
Deployment	Downtime during deployment	≤ 30 min
User Experience	Customer satisfaction score (post-UAT)	$\geq 8/10$
Maintenance	Mean time to resolve critical issues	≤ 4 hrs
Business Impact	Increase in online sales in 3 months	$\geq 25\%$

SoleMate – E-Commerce Application Project Evaluation Scoring Metrics

Total Score: 100 points (Weighting by SDLC phase)

1. Requirement Gathering & Analysis – 15 Points

Metric	Description	Max Points	Guideline
Clear project purpose	Purpose is defined and understood by stakeholders.	3	3 = Fully clear, 2 = Mostly clear, <2 = Unclear
Functional requirements completeness	Coverage of all required features/functions in documentation.	5	5 = All documented, 3–4 = Most documented, <3 = Missing details

Non-functional requirements	Coverage of performance, scalability, usability, security constraints.	3	3 = All key NFRs stated, 2 = Partial, <2 = Missing
Stakeholder validation & acceptance criteria	Validation of requirements and defined acceptance tests.	4	4 = Fully validated & measurable, 3 = Partial, <3 = None

2. Planning & Estimation – 10 Points

Metric	Description	Max Points	Guideline
Project roadmap/timeline	Clear schedule with milestones.	3	3 = Complete roadmap, 2 = Partial, <2 = None
Task breakdown & dependencies	Clear work breakdown structure (WBS) with dependencies.	3	3 = Detailed, 2 = Adequate, <2 = Weak
Estimation accuracy	Estimates closely match actual work required.	2	2 = Accurate, 1 = Moderate deviation, 0 = Poor estimation
Risk analysis & mitigation	Identification and handling of project risks.	2	2 = Strong, 1 = Partial, 0 = None

3. System Design – 15 Points

Metric	Description	Max Points	Guideline
Architecture diagrams & patterns	Existence and clarity of system diagrams and chosen patterns.	5	5 = Comprehensive, 3–4 = Adequate, <3 = Poor
Database schema & data flow	Logical/physical schema and planned data movement.	4	4 = Strong, 3 = Adequate, <3 = Weak
Technology stack rationale	Justification for selected tech stack and tools.	3	3 = Fully justified, 2 = Somewhat, <2 = None
Scalability & security in design	Handling of scale-out and security in architecture.	3	3 = Fully addressed, 2 = Partial, <2 = Not addressed

4. Development – 25 Points

Metric	Description	Max Points	Guideline
Code readability & maintainability	Clarity, comments, and ease of modifications.	5	5 = Excellent, 3–4 = Adequate, <3 = Poor
Modularity & separation of concerns	Logical, reusable, and decoupled modules.	5	5 = Excellent, 3–4 = Adequate, <3 = Weak

Adherence to coding standards & linting	Following style guides and automated lint checks.	5	5 = Fully compliant, 3–4 = Mostly, <3 = Poor
Error handling & logging	Graceful error capture and informative logging.	5	5 = Comprehensive, 3–4 = Adequate, <3 = Weak
Performance & efficiency	Code and algorithms optimised for runtime/resource usage.	5	5 = Optimised, 3–4 = Adequate, <3 = Poor

5. Testing – 15 Points

Metric	Description	Max Points	Guideline
Unit test coverage	Degree of coverage for individual functions/modules.	5	5 = >90%, 3–4 = 60–90%, <3 = <60%
Integration & e2e tests	Test coverage for multi-module and full flows.	5	5 = Comprehensive, 3–4 = Some, <3 = Minimal
Automated tests in CI/CD	Automated testing as part of pipeline.	3	3 = Fully automated, 2 = Partially, <2 = None
Performance/security testing	Includes non-functional testing.	2	2 = Both done, 1 = One done, 0 = None

6. Deployment – 10 Points

Metric	Description	Max Points	Guideline
Deployment documentation	Clarity of deployment steps.	2	2 = Clear, 1 = Partial, 0 = None
Automated deployment	Automatic deploy to environments.	3	3 = Full automation, 2 = Partial, <2 = None
Multi-environment support	Separate dev/stage/prod with parity.	3	3 = All present, 2 = Basic, <2 = None
Scalability, rollback, monitoring	Ops readiness at deployment.	2	2 = All in place, 1 = Partial, 0 = None

7. Documentation – 5 Points

Metric	Description	Max Points	Guideline
README quality	Completeness and clarity of README.	2	2 = Complete, 1 = Partial, 0 = None
Setup & usage guides	Step-by-step install and usage instructions.	2	2 = Clear, 1 = Partial, 0 = None

Troubleshooting & FAQs	Answers to common issues.	1	1 = Present, 0 = Absent
------------------------	---------------------------	---	-------------------------

8. Maintenance & Support – 5 Points

Metric	Description	Max Points	Guideline
Frequency of updates	How often releases/patches occur.	2	2 = Regular, 1 = Occasional, 0 = None
Issue tracking & resolution	Managing bug reports and turnaround time.	2	2 = Active & fast, 1 = Adequate, 0 = Poor
User feedback integration	Including feedback into product iteration.	1	1 = Present, 0 = Absent

Scoring Summary Table

SDLC Phase	Max Points
Requirement Gathering	15
Planning & Estimation	10
System Design	15
Development	25
Testing	15
Deployment	10
Documentation	5
Maintenance & Support	5
Total	100